

National University of Singapore
Department of Mathematics

QF4204 Final-Year Project

Monte Carlo vs Quasi-Monte Carlo for Path-Dependent Option Pricing

An Empirical Study of Convergence
and Variance Reduction Techniques

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Abstract

This study compares Monte Carlo (MC) and randomised quasi-Monte Carlo (RQMC) using independently scrambled Sobol' nets for pricing a discretely monitored arithmetic Asian call under the Black–Scholes model. Six estimators comprise the two baseline methods and their separate antithetic-variate and Gaussian importance-sampling variants. Ten contract configurations vary strike and monitoring frequency, using three matched payoff-evaluation budgets and 32 independent replicates per comparison. Each contract has an independent numerical benchmark and a separate importance-sampling pilot. Baseline RQMC achieves smaller measured root-mean-square error (RMSE) than baseline MC throughout the tested grid. At 64 monitoring dates and 16,384 payoff evaluations, RQMC reduces RMSE by factors of approximately 2.2–17.7 relative to MC. Variance-reduction gains depend on the contract. At the largest budget, MC antithetics are especially effective for in-the-money calls, while importance sampling reduces measured variance for both samplers at the highest tested strike but increases RQMC variance at lower strikes. These gains also vary with monitoring frequency. Runtime comparisons account for pilot costs and do not establish a universal ranking at equal computational cost. The findings support RQMC as a strong baseline while showing that additional variance reduction requires assessment for the specific contract and computational budget.

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1 Introduction

1.1 Background and motivation

Pricing a derivative requires accounting for uncertainty in the future value of its underlying asset. Under risk-neutral valuation, many option prices can be expressed as expectations of discounted payoffs. Although analytical formulas are available for some contracts, more complex payoff structures often require numerical methods. Monte Carlo (MC) simulation offers a flexible approach by estimating these expectations through the average of discounted payoffs across simulated paths. Its ability to accommodate different payoff structures and to simulate entire asset paths makes it particularly useful for path-dependent derivatives (Glasserman, 2003, Chapter 1).

Asian options provide a natural setting for studying this application. Their payoffs depend on an average of the underlying asset price over an observation period, making them relevant when an exposure is tied to prices over time rather than to a single terminal price. For an arithmetic Asian call, the payoff depends on whether the arithmetic average exceeds a fixed strike. Even under the Black–Scholes framework, this average does not generally have the lognormal distribution that permits the familiar European-call pricing formula. Consequently, arithmetic Asian options generally require numerical valuation or approximation, and simulation provides a direct way to incorporate their path dependence (Glasserman, 2003, Sections 1.1.2 and 3.2.2).

The flexibility of standard MC comes with a relatively slow reduction in sampling error. Let Y denote a discounted payoff with price $V = \mathbb{E}[Y]$ under the risk-neutral measure and finite variance $\sigma_Y^2 = \mathrm{Var}(Y)$. For N independent and identically distributed (IID) payoff samples, the standard MC estimate and its root-mean-square error (RMSE) satisfy

$$\widehat{V}_N = \frac{1}{N}\sum_{i=1}^{N} Y_i, \qquad \mathrm{RMSE}(\widehat{V}_N) = \sqrt{\mathbb{E}\left[(\widehat{V}_N - V)^2\right]} = \frac{\sigma_Y}{\sqrt{N}}. \tag{1}$$

Thus, the error decreases at the rate $O(N^{-1/2})$ (Owen, 2020, Section 1). Reducing this error by a factor of ten therefore requires one hundred times as many samples. Although this convergence exponent does not depend on the number of simulated inputs, the payoff variance and cost of generating each path can depend on the problem. Achieving high pricing accuracy can therefore become computationally expensive.

1.2 Quasi-Monte Carlo and Sobol' sequences

Quasi-Monte Carlo (QMC) methods offer an alternative by replacing independent pseudo-random sampling with deliberately structured points. Low-discrepancy sequences aim to cover the integration domain more evenly, reducing the gaps and clustering that arise in random samples. Sobol' sequences are an established construction for this purpose and form the basis of QMC in this project. Giles et al. (2008) explain their use in financial applications and the role of randomisation in allowing statistical error assessment. Deterministic QMC and randomised QMC (RQMC) are distinct, where the latter introduces randomness while retaining the structure of the point set. This study uses independently scrambled Sobol' replicates when assessing QMC sampling uncertainty.

Prior financial applications motivate the choice of Sobol' sequences, while also showing that the payoff and path construction affect the comparison. Section 2 reviews this evidence and explains the contribution of the present experimental design.

1.3 Dimensionality and variance reduction

Path dependence makes simulation dimensionality central to this comparison. With a single asset and a standard Brownian-increment construction, increasing the number of monitoring dates increases the number of random inputs needed to generate each path. However, the nominal number of inputs alone need not determine QMC performance, as some inputs or combinations of inputs may account for much more of the payoff variation than others. This distinction motivates the notion of effective dimension discussed by Giles et al. (2008). The project examines how performance changes with monitoring frequency, recognising that changing the number of monitoring dates also changes the Asian contract being priced.

Variance-reduction techniques provide a further means of improving simulation efficiency. This study focuses on antithetic variates (AV) and importance sampling (IS). The AV method pairs related simulated outcomes to reduce the variability of their average. IS changes the sampling distribution to give greater attention to influential outcomes, with compensating weights preserving the pricing expectation (Glasserman, 2003, Chapter 4). The latter is particularly relevant to out-of-the-money options, for which positive payoffs may occur infrequently under the original distribution. Moneyness therefore offers a useful way to investigate when these techniques are most effective. As Giles et al. (2008) discuss, techniques developed to improve MC can also benefit QMC. A direct comparison is therefore needed to determine whether these gains depend on moneyness and dimensionality.

1.4 Project aims and research questions

The aim of this project is to evaluate the convergence, pricing accuracy, and computational efficiency of standard MC and Sobol'-based QMC for a fixed-strike, discretely monitored arithmetic Asian call under the Black–Scholes framework. The study assesses AV and IS relative to their corresponding MC and QMC baselines, considering both statistical efficiency and computational cost. It is organised around four research questions:

1. How does QMC compare with standard MC when pricing an arithmetic Asian option?
2. How does the performance of MC and QMC change as the number of monitoring dates, and hence the simulation dimensionality, increases?
3. Do classical variance-reduction techniques improve the performance of QMC, and how does the magnitude of improvement compare with their effect on standard MC?
4. How does the effectiveness of variance-reduction techniques depend on option characteristics, particularly moneyness and simulation dimensionality?

2 Related Literature and Study Contribution

2.1 Evidence for QMC in financial valuation

The literature motivates QMC as an empirical alternative to MC, while also showing why performance must be assessed for a specified payoff and implementation. Paskov and Traub (1995) compare Sobol' and Halton constructions with ordinary and antithetic MC for a collateralised mortgage obligation (CMO) with ten tranches and integration dimensions up

to 360. Their Sobol' implementation performs favourably in that application, including at relatively small sample sizes. However, the authors explicitly limit the reported evidence to the tested CMO. Its payoff structure and interest-rate setting differ from the present Asian call, so its numerical speedups cannot be transferred directly. The relevant motivation is that high nominal dimension alone need not prevent useful QMC gains.

Acworth et al. (1998) examine option-pricing examples in which dimension arises from monitoring dates or multiple assets. They compare MC and antithetic MC with several low-discrepancy constructions, and examine alternative Brownian-path constructions. Their average-rate example uses a geometric average as a tractable surrogate for an arithmetic average. They assess root-mean-square relative error across randomly generated problem instances, whereas the present study calculates RMSE across independent randomisations of a fixed contract. Their results highlight the importance of path construction alongside the choice of point sequence in determining QMC performance. The present study holds the forward-increment construction fixed to examine the effects of sampling, AV and IS under a common mapping from inputs to paths. Brownian bridge and principal component analysis remain outside the scope of this study.

These studies support comparing QMC against both ordinary and modified MC. They do not establish that a technique which improves MC will improve an already effective QMC estimator by the same proportion. That distinction motivates reporting gains relative to each sampler's own baseline.

2.2 Randomisation, uncertainty and variance reduction

Giles et al. (2008) connect the practical use of QMC with effective dimension, point construction and randomisation. Their illustrative arithmetic Asian example involves multiple assets and a randomly shifted lattice rule, and shows how the arrangement of the driving variables affects performance. Their findings therefore provide methodological guidance rather than a direct numerical benchmark for the present single-asset scrambled Sobol' experiment. Their discussion of independent randomisations provides the basis for assessing uncertainty from complete replicate estimates. Independent randomisations allow MC and RQMC to be compared using replicate RMSE, estimator variance and the standard error of the replicate mean, without treating points within a Sobol' net as independent observations.

L'Ecuyer et al. (2023) examine confidence intervals constructed from independent RQMC replicates. Their numerical study finds favourable coverage for Student-t intervals on the tested problems, supporting their use as approximate uncertainty summaries rather than guaranteed finite-sample bounds.

Owen (1998) develops variance results for scrambled nets, including those arising from Sobol' sequences. The randomisation preserves net balance and gives each point a uniform marginal distribution, supporting unbiased integration while retaining dependence between points. Owen (2020) shows why dropping the first Sobol' point can destroy the net structure and worsen convergence. These findings motivate the use of complete power-of-two Sobol' nets and independent randomisations in the present study. The specific scrambling implementation is specified in Section 4.1. However, the convergence rates established in these papers are not assumed to apply directly to the transformed option payoff.

For variance reduction, Glasserman (2003, Chapter 4) provides the antithetic covariance argument and the classical IS framework. He et al. (2023) analyse RQMC IS with

Gaussian and Student-t proposals. They give examples in which a Gaussian Laplace proposal reduces MC variance while making the transformed integrand less favourable for QMC. This distinguishes the IID variance objective from the conditions governing RQMC convergence and motivates assessing IS against each sampler's own baseline. Chen et al. (2025) study RQMC with IS, deriving error bounds under smoothness, growth and proposal-tail conditions. Their financial experiments combine preintegration with optimal drift importance sampling, including arithmetic Asian calls under Black–Scholes. This provides a closer connection between IS and RQMC, but the present study uses an unsmoothed payoff and a scalar shift selected by an independent pilot. Its empirical gains therefore cannot be inferred from that paper's convergence results or numerical comparisons.

2.3 Asian-option comparisons and the present contribution

Hossen and Hossain (2021) provide a more directly related empirical comparison of ordinary MC, antithetic MC, and QMC using Halton and Sobol' sequences. Their arithmetic Asian examples vary strike price and the number of simulation steps, with prices and error plots used to compare performance. This supports studying moneyness and monitoring frequency together. However, their reference prices are not adopted in this study, as differences in monitoring conventions may affect the resulting option prices. The present study instead constructs separate reference prices, together with estimates of their uncertainty, for each combination of monitoring frequency and strike price.

Xu et al. (2023) combine Sobol'-based QMC with a control variate for arithmetic-average Asian option pricing, using GARCH(1,1) volatility. Their numerical comparisons report improved error performance from combining the two techniques. This provides further motivation for investigating variance reduction alongside QMC. However, their control-variate estimator and volatility model differ from the present antithetic and IS comparisons under constant-volatility Black–Scholes. Their findings therefore motivate the general combination without establishing the gains of the specific methods tested here.

The contribution is a controlled empirical comparison, rather than a new pricing model or a new low-discrepancy construction. All six methods share the same path simulation and payoff calculation, and AV and IS comparisons reuse the same baseline estimates for each contract specification. Complete power-of-two Sobol' nets, independent scrambles, and matched payoff evaluations ensure consistent comparisons across the different methods. Separate pilot samples and reported tuning costs allow the statistical gains from IS to be distinguished from its standalone computational cost. The comparison across strike prices and monitoring frequencies then examines whether these improvements remain consistent under different option characteristics. These choices allow the findings to be interpreted within a small, reproducible study without claiming a universal ranking or novelty for the underlying methods.

3 Methodology

This section specifies the pricing problem, baseline estimators, and the AV and IS modifications. The model, path construction, averaging convention, and discounting are kept consistent across methods. Unless otherwise stated, expectations and variances of estimators are taken with respect to their sampling distributions, while $\mathbb{E}^{\mathbb{Q}}$ denotes

expectation under the risk-neutral measure. Contract parameters other than d are suppressed in V_d and the payoff functions.

3.1 Model assumptions and option contract

The underlying is a single non-dividend-paying asset with initial price $S_0 > 0$. Under the risk-neutral measure $\mathbb{Q}$, its price follows the Black–Scholes dynamics

$$\mathrm{d}S_t = rS_t\,\mathrm{d}t + \sigma S_t\,\mathrm{d}W_t^{\mathbb{Q}}, \qquad 0 \le t \le T, \tag{2}$$

where r is the constant continuously compounded risk-free rate, $\sigma \ge 0$ is the constant volatility, $T > 0$ is the maturity, and $W^{\mathbb{Q}}$ is a standard Brownian motion. These assumptions provide a common framework for comparing numerical pricing performance (Chen et al., 2025, Section 5.1).

The contract is a fixed-strike arithmetic Asian call with strike $K > 0$ and d equally spaced monitoring dates $t_j = jT/d$, $j = 1, \ldots, d$. Its arithmetic average, terminal payoff, and time-zero price are

$$A_d = \frac{1}{d}\sum_{j=1}^{d} S_{t_j}, \qquad \varphi_d = (A_d - K)^+, \tag{3}$$

$$V_d = \mathbb{E}^{\mathbb{Q}}\left[e^{-rT}\varphi_d\right], \tag{4}$$

where $x^+ = \max(x, 0)$. The initial spot S_0 is excluded from the average, the maturity price is included, and the payoff is paid at T. The subscript d emphasises that a change in monitoring frequency changes the contract as well as the simulation dimension. Each monitoring schedule therefore requires its own reference price.

Moneyness is described using the strike-to-spot ratio K/S_0, with values below, equal to, and above one labelled as in, at, and out of the money, respectively. However, note that this classification is based on the initial asset price, whereas the actual payoff depends on whether the arithmetic average A_d exceeds the strike K. Therefore, the moneyness classification does not directly determine the likelihood of a positive payoff.

Two limiting cases provide useful checks of the contract definition. With $d = 1$, the average is S_T and the price reduces to that of a European call. With $\sigma = 0$, the path is deterministic and the price is

$$V_d = e^{-rT}\left(\frac{S_0}{d}\sum_{j=1}^{d} e^{rt_j} - K\right)^+. \tag{5}$$

3.2 Shared path construction and integration formulation

Let $\Delta t = T/d$. Given independent standard normal variables $Z_1, \ldots, Z_d$, the asset prices at the monitoring dates are generated by

$$S_{t_j} = S_{t_{j-1}} \exp\left[\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t}\,Z_j\right], \qquad j = 1, \ldots, d. \tag{6}$$

This recurrence relation uses the exact risk-neutral transition distribution, rather than a time-stepping approximation of the underlying stochastic differential equation (Chen

et al., 2025, Section 5.1, equations (17)–(19)). Consequently, the simulated prices do not introduce time-discretisation bias when valuing the specified discretely monitored Asian option. This differs from pricing a continuously monitored Asian option, where discretising the averaging period may introduce approximation error.

Write $g_d(\boldsymbol{z}) = e^{-rT}(A_d(\boldsymbol{z}) - K)^+$ for the discounted payoff obtained from the normal input vector $\boldsymbol{z}$, where $A_d(\boldsymbol{z})$ uses the path in (6). If Φ is the standard normal cumulative distribution function, define

$$f_d(\boldsymbol{u}) = g_d\Big(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_d)\Big),$$
$$V_d = \int_{(0,1)^d} f_d(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u}. \tag{7}$$

This expresses pricing as a d-dimensional integration problem to which both MC and QMC can be applied (Giles et al., 2008, Section 3). Although standard normal variables can be generated directly for MC, Sobol' sequences produce points in the unit hypercube. The inverse normal transformation converts these uniform inputs into the standard normal variables required for path simulation. Applying the same transformation to both MC and QMC ensures a consistent path construction, with the methods differing only in how their uniform inputs are generated.

A numerical consideration when applying this transformation is that the inverse normal CDF is finite only for inputs strictly between zero and one. Any endpoints produced in finite-precision computation must therefore be moved into the open interval using a documented numerical convention. Both implementations clip uniform coordinates to $[\varepsilon, 1-\varepsilon]$ with $\varepsilon = 2^{-53}$ and record how many coordinates are adjusted. The validation in Section 4.4 includes a sensitivity check with $\varepsilon = 2^{-40}$. Endpoint adjustment can introduce numerical bias and is distinct from the absence of time-discretisation bias in the path recursion. Non-finite transformed values or payoffs must be identified explicitly, without discarding samples. The distributional identities below describe the ideal estimators before finite-precision modifications.

3.3 Baseline Monte Carlo estimator

For one MC replicate, generate N independent uniform vectors $\boldsymbol{U}_i$ on $(0,1)^d$ and evaluate $Y_i = f_d(\boldsymbol{U}_i)$. The estimator is

$$\widehat{V}_{N,d}^{\mathrm{MC}} = \frac{1}{N}\sum_{i=1}^{N} Y_i, \qquad \mathbb{E}\Big[\widehat{V}_{N,d}^{\mathrm{MC}}\Big] = V_d. \tag{8}$$

Since the discounted payoff has finite variance under this model, the sampling-error relation in (1) applies. For $N > 1$, the payoff sample variance and the estimated standard error of this single replicate are

$$s_Y^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i - \widehat{V}_{N,d}^{\mathrm{MC}})^2, \qquad \widehat{\mathrm{SE}}\Big(\widehat{V}_{N,d}^{\mathrm{MC}}\Big) = \frac{s_Y}{\sqrt{N}}. \tag{9}$$

This calculation relies on independent payoffs (Glasserman, 2003, Section 1.1). While this provides a measure of uncertainty for a single MC estimate, comparisons across methods are instead based on independent replicate estimates, as discussed in the following sections.

3.4 Baseline scrambled Sobol' estimator

For QMC, replace the independent uniform vectors by a Sobol' point set of size $N = 2^m$, where m is a nonnegative integer. Each replicate uses a complete net, preserving its balance properties by avoiding arbitrary skipping, thinning, or truncation. Sobol' nets and their uniform coverage are described by Giles et al. (2008, Section 2), and the importance of retaining complete nets rather than dropping the first point is examined by Owen (2020).

For statistical comparisons, apply an independent random scramble to each replicate. The randomisation must preserve the net structure and make each point marginally uniform on the integration domain. Denoting the points in replicate ℓ by $\boldsymbol{U}_{\ell i}^{\mathrm{RQMC}}$, the estimate is

$$\widehat{V}_{N,d}^{\mathrm{RQMC},(\ell)} = \frac{1}{N}\sum_{i=1}^{N} f_d(\boldsymbol{U}_{\ell i}^{\mathrm{RQMC}}). \tag{10}$$

RQMC is the form of QMC used in the statistical comparisons throughout this report. Under ideal randomisation, each Sobol' point is uniformly distributed, ensuring that the resulting price estimator is unbiased (Owen, 1998, Section 3). However, unlike standard MC, the points within each Sobol' net are not independent. As a result, the standard error formula in (9), which assumes independent payoff samples, cannot be applied directly. Uncertainty must instead be estimated from independent randomisations of the whole net (Giles et al., 2008; Owen, 2020). All QMC results reported in this study use independently scrambled Sobol' nets.

3.5 Replication and computational-budget notation

Let M denote a method and let $\widehat{V}_{N,d}^{M,(1)}, \ldots, \widehat{V}_{N,d}^{M,(R)}$ be $R \geq 2$ independent replicate estimates at a fixed contract configuration and sample size. MC replicates use independent random streams, while RQMC replicates use independent scrambles. The reported mean and the estimated variance of a single replicate estimator are

$$\overline{V}_{N,d}^{M} = \frac{1}{R}\sum_{\ell=1}^{R} \widehat{V}_{N,d}^{M,(\ell)}, \tag{11}$$

$$s_M^2 = \frac{1}{R-1}\sum_{\ell=1}^{R}\left(\widehat{V}_{N,d}^{M,(\ell)} - \overline{V}_{N,d}^{M}\right)^2,$$

$$\widehat{\mathrm{SE}}(\overline{V}_{N,d}^{M}) = \frac{s_M}{\sqrt{R}}. \tag{12}$$

Thus s_M^2 describes variability between replicate price estimates, whereas $s_M/\sqrt{R}$ describes uncertainty in their reported mean. Neither is the variance of individual payoffs within a Sobol' net. This use of independent randomisations for variance and standard-error estimation follows the replication approach described by L'Ecuyer et al. (2023, Section 3). The empirical RMSE relative to an independently generated numerical benchmark V_d^{ref} is

$$\widehat{\mathrm{RMSE}}_M = \left[\frac{1}{R}\sum_{\ell=1}^{R}\left(\widehat{V}_{N,d}^{M,(\ell)} - V_d^{\mathrm{ref}}\right)^2\right]^{1/2}. \tag{13}$$

This measures the pricing error of individual replicate estimates. The absolute pricing error of the reported mean is $|\overline{V}_{N,d}^{M} - V_d^{\mathrm{ref}}|$, which is calculated using all R replicates and is distinct from the replicate RMSE. The contract-specific benchmarks are described in

Section 4.2. To ensure meaningful comparisons, the uncertainty in these benchmark prices should be small relative to the measured pricing errors.

For comparisons of a reported mean with an independent reference, the estimated standard error of their difference is $\sqrt{\widehat{\mathrm{SE}}(\overline{V}_{N,d}^{M})^2 + \widehat{\mathrm{SE}}(V_d^{\mathrm{ref}})^2}$. Agreement within this estimated uncertainty provides evidence that the results are consistent with the reference price, although it does not establish that the estimator is unbiased.

Throughout the study, d denotes the number of monitoring dates, N the number of base input vectors per replicate, R the number of independent replicates, and B the number of payoff evaluations per replicate. To ensure fair comparisons, all methods use the same payoff evaluation budget B, although the number of base input vectors N may differ between methods. The table below summarises these counts, with the additional evaluations required for IS tuning accounted for separately.

Method	Base vectors N	Pairs	Payoff evaluations
MC / RQMC	B	0	B
MC / RQMC with antithetics	$B/2$	$B/2$	B
MC / RQMC with importance sampling	B	0	B

The mean estimate is calculated from R independent replicates, each using B payoff evaluations, giving a total of RB evaluations. All methods are compared using the same values of B and R, with both pricing accuracy and runtime reported.

Choosing B as a power-of-two, with $B \geq 2$, permits complete Sobol' nets for both the baseline and antithetic methods, using $N = B$ and $N = B/2$, respectively. Convergence plots use the payoff evaluation budget B on the horizontal axis, with the number of replicates R also indicated. Numerical parameter values and replication counts are specified in the experimental design.

3.6 Antithetic variates

Antithetic sampling uses the symmetry of the standard normal distribution. For each base vector $\boldsymbol{Z}_i$, construct one path from $\boldsymbol{Z}_i$ and another from $-\boldsymbol{Z}_i$. Define the discounted pair average and the resulting estimator by

$$P_i = \frac{g_d(\boldsymbol{Z}_i) + g_d(-\boldsymbol{Z}_i)}{2}, \qquad \widehat{V}_{N,d}^{M+\mathrm{AV}} = \frac{1}{N}\sum_{i=1}^{N} P_i, \tag{14}$$

where M is MC or RQMC and the replicate index is suppressed. Both members of a pair have the correct marginal distribution, so their average retains expectation V_d. The second path is generated by negating every component of the original normal input vector, while keeping the asset price dynamics, payoff calculation, and discounting unchanged (Glasserman, 2003, Section 4.2).

For IID MC, write $\sigma_Y^2 = \mathrm{Var}(g_d(\boldsymbol{Z}))$ and $c_d = \mathrm{Cov}(g_d(\boldsymbol{Z}), g_d(-\boldsymbol{Z}))$. At the matched budget of $B = 2N$ payoff evaluations,

$$\mathrm{Var}(\widehat{V}_{N,d}^{\mathrm{MC+AV}}) = \frac{\sigma_Y^2 + c_d}{2N}, \qquad \mathrm{Var}(\widehat{V}_{2N,d}^{\mathrm{MC}}) = \frac{\sigma_Y^2}{2N}. \tag{15}$$

Thus negative covariance gives a variance reduction at equal payoff cost. For the arithmetic Asian call with the increment construction in (6), g_d is nondecreasing in each normal

input, whereas $g_d(-\boldsymbol{z})$ is nonincreasing. This monotonicity under independent normal coordinates gives nonpositive covariance, although the magnitude of the benefit depends on the contract parameters. The general monotonicity argument is discussed by Glasserman (2003, Section 4.2).

The independent observations for MC uncertainty are the N pair averages. For $N > 1$, let s_P be their sample standard deviation, and the estimated standard error of one estimator is $s_P/\sqrt{N}$. The $2N$ individual payoffs cannot be treated as independent because the two payoffs within each pair are dependent. For RQMC, each replicate uses a complete scrambled Sobol' net containing $N = 2^m$ base points, with antithetic pairing applied to the corresponding normal vectors as defined in (14). Unlike standard MC, RQMC variance is influenced not only by the variability of individual payoffs but also by the dependence between points within the Sobol' net. Consequently, a technique that reduces variance under IID sampling may not provide the same benefit for RQMC, and the variance reduction shown in (15) does not necessarily apply. Its benefit is therefore assessed empirically at equal payoff budgets, with uncertainty calculated across independent scrambles using (12).

The Sobol' construction can also be understood as applying the unchanged N-point base net to a symmetrised integrand. With f_d defined in (7), let

$$h_d(\boldsymbol{u}) = \frac{f_d(\boldsymbol{u}) + f_d(\mathbf{1} - \boldsymbol{u})}{2}. \tag{16}$$

The substitution $\boldsymbol{v} = \mathbf{1} - \boldsymbol{u}$ shows that $\int h_d(\boldsymbol{u})\,\mathrm{d}\boldsymbol{u} = V_d$, so symmetrisation preserves the pricing expectation. The identity $\Phi^{-1}(1-u) = -\Phi^{-1}(u)$ shows that reflecting the uniform inputs is equivalent to negating the corresponding normal inputs. The integration dimension remains d, and no base points are skipped or thinned. Note that the combined original and reflected points need not equal the standard $2N$-point Sobol' construction, and that equivalence is not required to apply the base net to h_d.

The implementation first transforms the uniform inputs into normal variables using the endpoint adjustment described in Section 3.2, before negating these variables to generate the antithetic paths. At fixed payoff budget, the comparison is between B base points applied to f_d and $B/2$ base points applied to h_d. Any benefit from symmetrisation must therefore compensate for the smaller base net. The comparison therefore asks whether spending the same payoff budget on additional Sobol' base points or on antithetic partners gives the more accurate estimator, with runtime reported separately.

3.7 Importance sampling

IS changes the distribution of the normal inputs to generate valuable payoff outcomes more frequently. A likelihood ratio compensates for this change so that the pricing expectation is preserved. Using the shifted Gaussian proposal family described by Chen et al. (2025, Section 2.1), let $\boldsymbol{X} = \boldsymbol{Z} + \boldsymbol{\theta}$, where $\boldsymbol{Z} \sim \mathcal{N}(\mathbf{0}, I_d)$, I_d is the d-dimensional identity matrix, and $\boldsymbol{\theta}$ is a fixed shift. If p is the standard normal density and $q_{\boldsymbol{\theta}}$ is the density of $\mathcal{N}(\boldsymbol{\theta}, I_d)$, their ratio is

$$L_{\boldsymbol{\theta}}(\boldsymbol{x}) = \frac{p(\boldsymbol{x})}{q_{\boldsymbol{\theta}}(\boldsymbol{x})} = \exp\!\left(-\boldsymbol{\theta}^{\mathsf{T}}\boldsymbol{x} + \frac{1}{2}\|\boldsymbol{\theta}\|^2\right), \tag{17}$$

where $\|\cdot\|$ denotes the Euclidean norm. The proposal density is positive everywhere, so it covers every normal input that can contribute to the price. Changing the integration

density gives

$$V_d = \int_{\mathbb{R}^d} g_d(\boldsymbol{x}) p(\boldsymbol{x}) \, \mathrm{d}\boldsymbol{x} = \mathbb{E}_{q_{\boldsymbol{\theta}}}[g_d(\boldsymbol{X}) L_{\boldsymbol{\theta}}(\boldsymbol{X})] \, . \tag{18}$$

This is the change-of-density identity in Chen et al. (2025, Section 2.1, equations (3)–(4)), applied to the discounted Asian payoff and a Gaussian mean shift. In terms of the original standard normal vector, the weighted discounted payoff is therefore

$$\begin{aligned} h_{d,\boldsymbol{\theta}}(\boldsymbol{z}) &= g_d(\boldsymbol{z} + \boldsymbol{\theta}) \exp\Big(-\boldsymbol{\theta}^{\mathsf{T}} \boldsymbol{z} - \frac{1}{2} \|\boldsymbol{\theta}\|^2 \Big) , \\ \widehat{V}_{N,d}^{M+\mathrm{IS}} &= \frac{1}{N} \sum_{i=1}^{N} h_{d,\boldsymbol{\theta}}(\boldsymbol{Z}_i). \end{aligned} \tag{19}$$

The shifted inputs generate paths under the proposal distribution using (6), and the likelihood ratio restores the risk-neutral expectation. The model parameters and discount factor are unchanged. The likelihood ratio weights each discounted payoff directly, without normalising by the sum of weights. At $\boldsymbol{\theta} = \mathbf{0}$, the weight is one and the estimator recovers the corresponding baseline for the same base points.

For every fixed finite shift in this model, the weighted payoffs have finite second moment, although an unsuitable shift can increase variance substantially. For MC, the weighted payoffs are independent and, for $N > 1$, their sample standard deviation divided by $\sqrt{N}$ estimates the standard error of a single replicate. For RQMC, the $\boldsymbol{Z}_i$ are inverse-normal transforms of a complete scrambled Sobol' net. The shift and weighting modify the integrand while preserving the underlying net; uncertainty is again estimated across independent scrambles. Both IS variants use $B = N$ payoff evaluations per evaluation replicate.

3.7.1 Shift selection and separation of tuning from evaluation

To keep tuning manageable and simple, this study uses the one-parameter family

$$\boldsymbol{\theta}(a) = \frac{a}{\sqrt{d}} \mathbf{1}_d, \qquad a \geq 0, \tag{20}$$

where $\mathbf{1}_d$ is the vector of ones. Each standard normal input is shifted by $a/\sqrt{d}$, encouraging higher asset paths and positive call payoffs when $a > 0$. Since $\|\boldsymbol{\theta}(a)\| = a$, increasing the dimension does not automatically increase the total shift magnitude. Using the same underlying normal inputs, let $S_T^{(a)}$ and $S_T^{(0)}$ denote the terminal asset prices with and without the shift. Then

$$\ln S_T^{(a)} - \ln S_T^{(0)} = \sigma \sqrt{\frac{T}{d}} \sum_{j=1}^{d} \frac{a}{\sqrt{d}} = \sigma a \sqrt{T}. \tag{21}$$

Thus, for fixed a, the additional terminal log-price displacement is independent of d. This shift specification is a simple modelling choice for the study, and is not intended to represent an optimal shift vector, unlike the optimal drift construction in Chen et al. (2025, Section 2.1, equation (5)), which selects a shift by maximising the product of the original normal density and the integrand.

For each contract configuration, an independent IID MC pilot compares a small, prespecified grid $\mathcal{A}$ of shift magnitudes including zero. The same n_{p} pilot normal vectors are used across candidates. Candidates that produce no positive payoffs during the pilot

simulations are excluded from the optimisation, since an estimated second moment of zero does not necessarily indicate low variance. Let $\mathcal{A}_+$ denote the remaining candidates. When $\mathcal{A}_+$ is nonempty, choose

$$\widehat{a} \in \arg\min_{a \in \mathcal{A}_+} \frac{1}{n_{\mathrm{p}}} \sum_{i=1}^{n_{\mathrm{p}}} h_{d,\boldsymbol{\theta}(a)}(\boldsymbol{Z}_i^{\mathrm{p}})^2. \tag{22}$$

The criterion estimates the second moment of the weighted payoff. Because its mean is V_d for every fixed shift, minimising its population second moment also minimises its IID variance (Glasserman, 2003, Section 4.6.1). Pilot estimates can nevertheless be unreliable, particularly when positive payoffs are rare. The number of positive payoffs for every candidate is therefore recorded. Excluding candidates with no positive payoffs helps prevent misleading estimates but does not guarantee reliable parameter selection. Ties are resolved in favour of the smaller shift. If $\mathcal{A}_+$ is empty, the pilot is flagged as uninformative and the zero-shift fallback is used. Numerical underflow, overflow or non-finite weights or weighted payoffs raise an error and stop the calculation. Such failures are not handled by excluding candidates and continuing the optimisation.

The selected shift is fixed before evaluation and shared by MC and RQMC for each contract configuration, across all evaluation budgets and replicates. All evaluation streams and scrambles are independent of the pilot. The estimator remains unbiased conditional on the pilot, because the pricing identity holds for every selected finite shift. Hence, the reported uncertainty reflects variation across evaluation replicates, with the chosen shift held fixed. The IID pilot criterion does not necessarily minimise RQMC variance, which is also affected by dependence between evaluations within the scrambled net.

The candidate grid, pilot size, seeds, selected shift, and pilot diagnostics are recorded with the experiment configuration. Each candidate requires n_{p} payoff evaluations, so tuning uses $n_{\mathrm{p}}|\mathcal{A}|$ evaluations in addition to the evaluation budget. The experimental settings and treatment of tuning cost are specified in Section 4.3. Checks of (17), zero-shift recovery, and finite weighted payoffs precede interpretation of IS results.

3.8 Comparison of variance-reduction gains

The core comparisons comprise MC, MC with AV, MC with IS, and the corresponding three RQMC methods. AV and IS are assessed separately. For each modification, define the estimated variance-reduction factor (VRF) at matched payoff budget B and replication count R by

$$\widehat{\mathrm{VRF}} = \frac{s^2_{\mathrm{baseline}}}{s^2_{\mathrm{modified}}}, \tag{23}$$

where both variances are computed from replicate price estimates using (12). The baseline is MC for an MC modification and RQMC for an RQMC modification. Values larger than 1 indicate a reduction in estimator variance, and values smaller than 1 indicate an increase. If the estimated denominator is zero, the ratio is flagged as undefined rather than reported as a finite gain. Accuracy and runtime accompany this measure, since variance reduction alone does not establish computational efficiency. Comparing these gains across strikes and monitoring dimensions addresses how the usefulness of each technique depends on the option characteristics.

4 Experimental Design and Validation

4.1 Contract grid and matched evaluation budgets

The main experiment compares all six methods across ten unique arithmetic Asian call contracts. The moneyness sweep uses $K \in \{90, 100, 110, 120\}$ at $d = 64$. A crossed comparison uses $K \in \{90, 100, 120\}$ at $d \in \{16, 64, 128\}$. The three overlapping contracts are evaluated once and reused in both comparisons. All other parameters remain fixed: $S_0 = 100$, $T = 1$, $r = 0.05$ and $\sigma = 0.20$, with no dividends. Moneyness is defined relative to the initial spot, while the option payoff depends on the discretely monitored arithmetic average.

Each configuration uses $B \in \{1\,024, 4\,096, 16\,384\}$ evaluation payoffs and $R = 32$ independent replicates per method and budget. For AV, the numbers of base vectors and pairs are both $B/2$, while other methods use B base vectors. Every RQMC replicate uses a complete independently scrambled Sobol' net, without skipping or thinning.[1] Each modified method is compared with its corresponding baseline at the same contract and budget. The baseline estimates are shared between the AV and IS comparisons. Pilot, benchmark and evaluation streams are separate across all contracts. Appendix A provides the complete experimental procedure, while the estimator definitions and comparison rules remain in the main text.

Changing d changes both the integration dimension and the discrete arithmetic average, as shown in Equation 3. Each monitoring schedule therefore has its own benchmark. The dimension comparison thus examines how performance changes with monitoring frequency, rather than isolating the effect of integration dimension alone.

4.2 Numerical benchmarks and precision

A benchmark is considered sufficiently precise when its estimated standard error (SE) is at most 10% of the measured replicate RMSE. This criterion aims to keep benchmark uncertainty small relative to the pricing errors being evaluated, reducing its potential influence on the measured RMSE.

Each reference is initially calculated as the mean of 16 independent baseline RQMC estimates, each using $2^{18} = 262\,144$ paths, for a total of $4\,194\,304$ payoff evaluations. The standard error is calculated across the 16 estimates. The initial reference for $K = 120, d = 128$ exceeded the 10% threshold for the two largest-budget IS comparisons, with its SE reaching 12.9% of the smallest measured RMSE. An additional 48 independent reference replicates were therefore appended, giving 64 scrambles and $16\,777\,216$ benchmark payoffs for this contract. All original evaluation estimates were retained. This refinement was made to improve reference precision, without changing the evaluation methods. All 180 comparisons (10 contracts $\times$ 6 methods $\times$ 3 budgets) then satisfied the 10% precision criterion, with the largest reference-SE-to-RMSE ratio being 9.04%. Benchmark inputs are separate from tuning and evaluation inputs. Table 1 reports the reference prices and IS shifts used in the evaluation.

This diagnostic is an empirical precision criterion, not a bound on reference error. The numerical reference shares the pricing implementation, making the independent mathematical validation in Section 4.4 essential.

[1] SciPy's Sobol' generator uses a left linear matrix scramble followed by a digital random shift (The SciPy Community).

Table 1: Contract-specific numerical references, their estimated standard errors, and pilot-selected scalar IS shifts.

K	d	Reference price	Reference SE	Shift $\widehat{a}$
90	16	12.838930	0.000127	0.5
100	16	6.058025	0.000280	1
120	16	0.635857	0.000192	2
90	64	12.656697	0.000286	0.5
100	64	5.836283	0.000536	1
110	64	2.045015	0.000479	1.5
120	64	0.552934	0.000458	1.5
90	128	12.626093	0.000327	0.5
100	128	5.799998	0.000449	1
120	128	0.540198	0.000289	1.5

4.3 Importance-sampling tuning and computational cost

Each contract has a separate IID MC pilot of 4,096 base vectors and the candidate grid $\mathcal{A} = \{0, 0.5, 1, 1.5, 2, 2.5, 3\}$. Common inputs are used across candidates, following the selection rule in Section 3.7. The resulting scalar shift is shared by MC+IS and RQMC+IS at every budget and replicate for that contract. Reported IS uncertainty therefore conditions on this one selected shift and excludes variability from repeating the tuning procedure.

Each pilot costs $4\,096 \times 7 = 28\,672$ payoff evaluations. Comparisons at fixed B match evaluation work, while standalone IS cost adds one full pilot to one evaluation. This gives $B + 28\,672$ payoffs and pilot time plus mean evaluation time. At the largest budget, this is 45,056 payoffs for IS versus 16,384 for the other methods. The actual experiment shares one pilot across both IS methods and all their budgets and replicates, and standalone costs must not be summed to obtain the experiment's total cost.

Runtime includes sampling, inverse normal transformation, path generation, payoff calculation, and estimation. IS pilot timing includes candidate evaluation and selection. Imports, file writing, figure generation and benchmark construction are excluded from evaluation timings. The reference cost is separate from the cost of a pricing method. Timing comparisons are specific to this implementation and execution environment.

4.4 Completed validation checks

Validation checks the pricing identities independently of the contract-specific numerical benchmarks. The principal checks and outcomes are summarised below.

- **Paths and payoffs.** Simulated prices agree with the closed-form solution for the underlying asset price for the same normal inputs. A separate payoff example confirms that the initial spot is excluded from the arithmetic average and that the payoff is discounted exactly once. With zero volatility, baseline and antithetic estimates reproduce the analytical deterministic price in (5) across three rates and three strikes.

- **Analytical price.** With one monitoring date, all six methods agree with the

analytical European-call price within the prescribed test tolerance. Table 2 gives the numerical evidence.

- **Antithetic pairing and uncertainty.** Independent path calculations reproduce the pair averages and the MC standard error based on those pairs. Replicate variance, standard error and RMSE also agree with a small known example.

- **Importance-sampling weights.** Weights agree with a direct ratio of normal densities, and weighted payoffs agree with independently constructed paths. A zero shift recovers the baseline price and standard error exactly for the same inputs. With zero volatility and one date, numerical integration at two fixed nonzero shifts, $a = 0.5$ and $a = 1.5$, recovers the same deterministic price. These two values of a are chosen manually, rather than through pilot tuning, to test the weighting at different shift magnitudes. Although individual weighted samples vary, the likelihood ratio has expectation one under the proposal, preserving the constant payoff's expectation.

Endpoint clipping ensures finite values after the inverse normal transformation. A selected 4,096-point, eight-dimensional scrambled net gives the same price under clipping thresholds 2^{-53} and 2^{-40}. However, this check does not establish a bound on clipping bias. No uniform inputs required endpoint clipping during the main experiments. Supplementary checks of sampling, pilot safeguards, reproducibility and cost accounting are provided in Appendix D. Variance reduction is not required for validation, as a correctly implemented estimator may still be less efficient than its baseline.

4.4.1 Numerical evidence from analytical cases

Table 2 makes the European-call check explicit. It uses $S_0 = K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$ and $d = 1$, so the analytical reference is 10.4505835722. Each method uses 16 independent replicate estimates at $B = 16\,384$ payoffs per replicate. The IS shift is fixed at $a = 1$ to test the nonzero-shift likelihood-ratio implementation. This value is chosen for validation rather than through pilot tuning, and is not claimed to be optimal. The SE column describes the mean of the 16 estimates and, for RQMC, is calculated across independent scrambles. No within-net payoff standard errors are used for RQMC.

Table 2: European-call validation against the analytical value 10.4505835722. Absolute error is measured for the replicate mean.

Method	Mean estimate	SE of mean	Absolute error
MC	10.44266075	0.02667557	0.00792282
RQMC	10.45074484	0.00018816	0.00016127
MC + AV	10.44710668	0.01873768	0.00347689
RQMC + AV	10.45067509	0.00020293	0.00009152
MC + IS	10.44738313	0.01202272	0.00320044
RQMC + IS	10.45057882	0.00000481	0.00000475

All six discrepancies are smaller than one estimated SE of their respective means in this check, and hence satisfy the prespecified tolerance of six estimated standard errors. Note that this threshold is an implementation check rather than a confidence interval. The same list of replicate seeds is reused across methods for this validation, although

replicates within each method are independent. These rows therefore confirm agreement with the analytical price but do not replace the main experiments, which use separate random seeds. In particular, the small RQMC+IS error in this one-dimensional example should not be used to predict its ranking for a high-dimensional Asian payoff.

Table 3 checks the arithmetic average itself. It sets $\sigma = 0$, $S_0 = 100$, $r = 0.05$, $T = 1$ and $d = 16$, with three strikes producing a large positive payoff, a small positive payoff and a zero payoff. Each method uses 256 payoff evaluations. The analytical value comes from prices at the absolute monitoring times in (5), independently of the simulated path recursion. The table shows the MC estimate and the largest absolute discrepancy across all six methods. All six agree at the displayed precision.

Table 3: Zero-volatility validation. The maximum error is taken across all six methods. IS uses zero shift in this deterministic check.

K	Analytical value	MC estimate	Maximum error
80	21.59528447	21.59528447	0
100	2.57069598	2.57069598	4.44×10^{-16}
120	0.00000000	0.00000000	0

The largest discrepancy is approximately 4.44×10^{-16}, consistent with floating-point rounding. There is no sampling uncertainty for this zero-shift deterministic payoff, so no estimated confidence interval is needed. A nonzero IS shift would introduce random likelihood weights even when the unweighted payoff is constant, as verified by the importance-sampling validation check. Together, the two tables validate the simulated European-call price against its analytical value and the arithmetic Asian average against a deterministic benchmark. However, these tests do not fully validate the methods in higher dimensions, so separate checks of IS weights, antithetic pairing and sampling remain necessary.

5 Results and Discussion

5.1 Accuracy checks and baseline convergence

The sweep contains 5,760 evaluation estimates across 10 contracts, 6 methods, 3 budgets and 32 independent replicates. All 180 method–contract–budget comparisons pass the benchmark-precision criterion after the reference refinement described in Section 4.2. No normal-transform endpoint adjustments or non-finite estimates occurred. At the largest budget, the maximum absolute difference between a method's replicate mean and its reference is 1.90 combined estimated standard errors. This agreement supports consistency of the methods with the numerical references. Figure C.1 in Appendix C illustrates the mean-price uncertainty for the $K = 120, d = 64$ contract at this budget.

Figure 1 compares convergence across strikes at $d = 64$. Baseline RQMC has smaller measured RMSE than baseline MC for all four strikes and all three budgets. At $B = 16\,384$, MC-to-RQMC RMSE ratios are 17.73, 6.89, 3.76 and 2.16 for strikes 90, 100, 110 and 120, respectively. Thus the relative RQMC advantage becomes smaller as the strike increases in this comparison, even though RQMC remains more accurate at the matched budget. A smaller absolute error for a higher-strike option does not necessarily imply better relative

accuracy, since higher-strike options generally have lower option prices, potentially leading to a larger relative error.

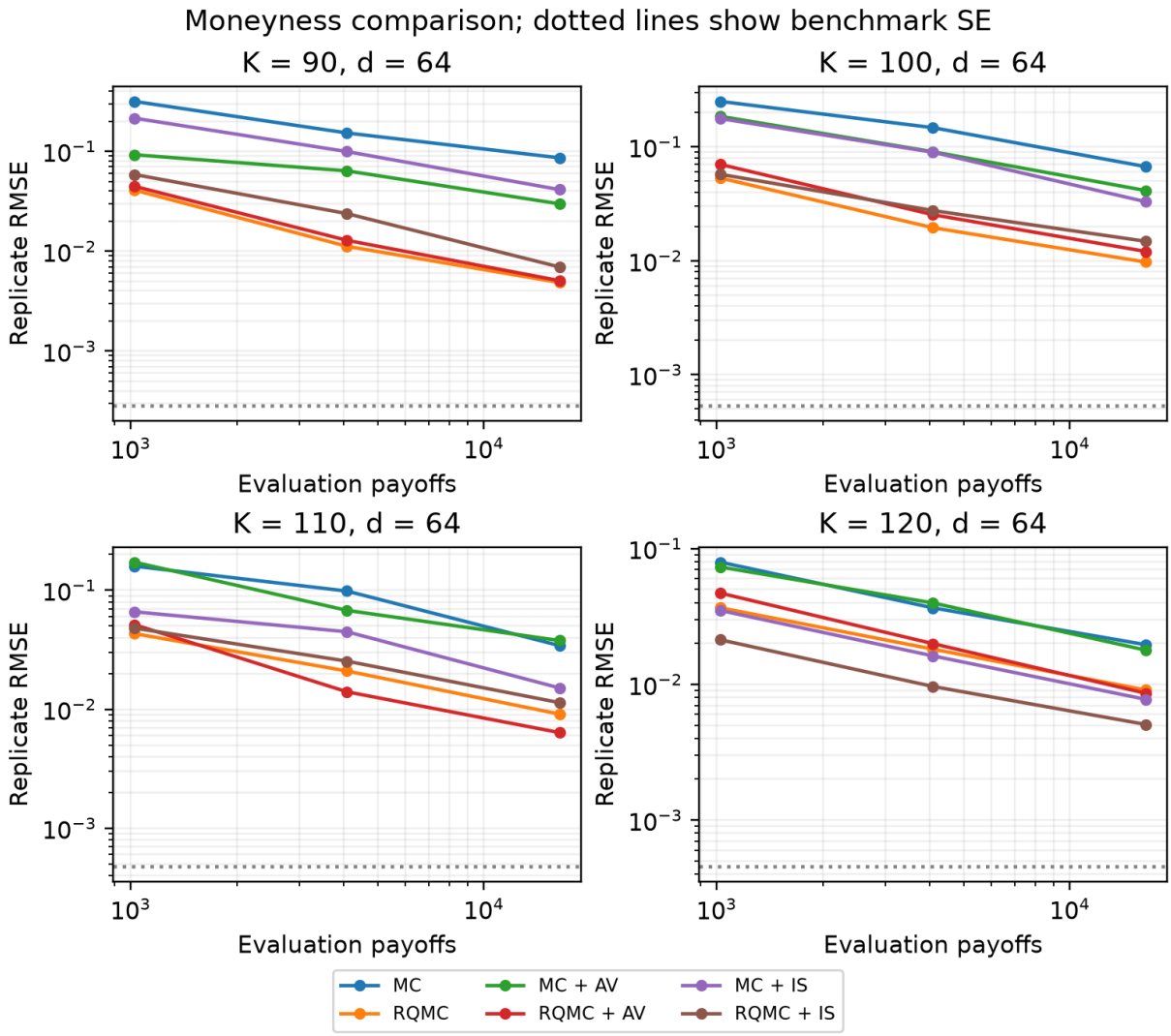


Figure 1: Replicate RMSE across four strikes at 64 monitoring dates. Each point uses 32 independent estimates at the stated evaluation budget, and dotted lines show the contract-specific reference SE. Lines connect measured points and do not represent fitted convergence rates.

Increasing the budget from 1,024 to 16,384 multiplies evaluation work by 16. At $d = 64$, measured baseline MC RMSE falls by factors between 3.67 and 4.66, consistent with the expected factor of four under inverse-square-root scaling. Baseline RQMC RMSE falls by factors between 4.04 and 8.43. These observations support a practical RQMC advantage over the tested range, although the results are insufficient to determine its asymptotic convergence rate. Table 6 in Appendix B gives largest-budget RMSE values for all ten configurations.

5.2 Variance reduction across moneyness

The following comparisons use estimated VRFs at the largest evaluation budget, $B = 16\,384$, with $R = 32$ independent replicates. Each modified estimator is compared with its

corresponding baseline, as defined in Section 3.8. Values close to 1 should be interpreted cautiously, since the factors are estimated from only 32 replicates. Figure 2 shows the moneyness comparison at $d = 64$, with MC in the left panel and RQMC in the right panel. Table 4 also reports results for $d = 16$ and $d = 128$.

Table 4: Estimated VRFs at $B = 16\,384$ and $R = 32$. Each modified estimator is compared with its own baseline. IS estimates use the pilot-selected shift, held fixed across replicates.

K	d	MC+AV	RQMC+AV	MC+IS	RQMC+IS
90	16	5.36	1.00	2.65	0.40
100	16	2.71	0.56	4.14	0.30
120	16	1.83	0.79	6.03	2.05
90	64	8.39	0.89	4.42	0.47
100	64	2.70	0.66	4.01	0.42
110	64	0.85	1.93	5.16	0.60
120	64	1.17	1.11	6.55	3.19
90	128	9.15	1.26	2.62	0.34
100	128	2.31	0.20	1.53	0.19
120	128	1.39	1.29	14.00	5.21

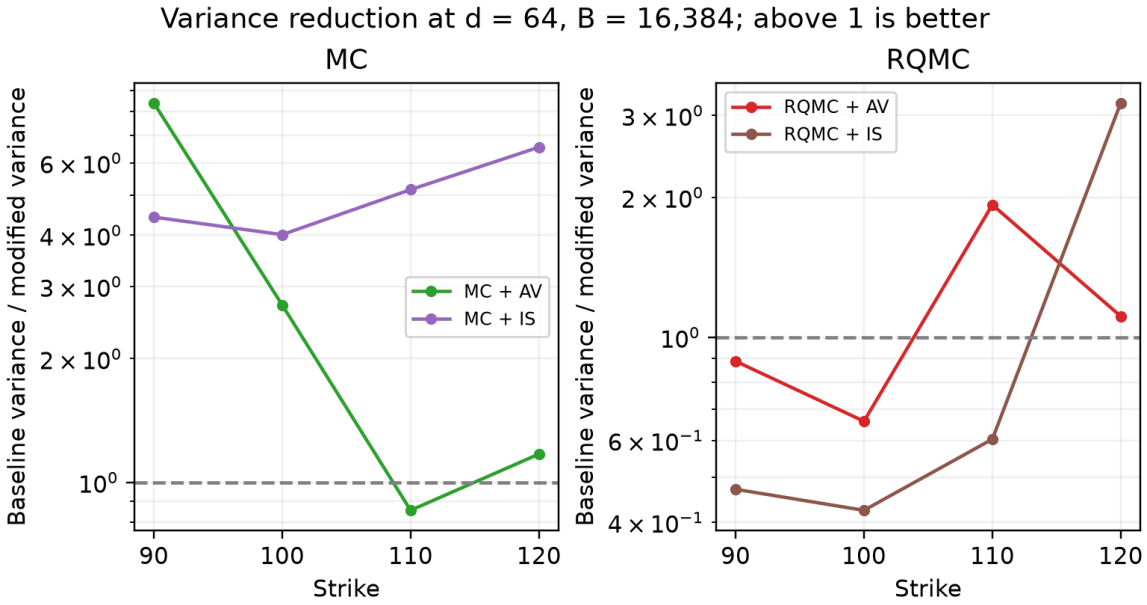


Figure 2: VRFs across strikes at $d = 64$ and $B = 16\,384$. Dashed lines mark no variance change.

For MC+AV at $d = 64$, the largest VRF is 8.39 at $K = 90$, followed by 2.70 at $K = 100$. The VRFs of 0.85 and 1.17 at $K = 110$ and $K = 120$ suggest little improvement for these more out-of-the-money options. The broader pattern of stronger gains at lower strikes also appears at $d = 16$ and $d = 128$. For lower strikes, both antithetic paths are more likely to produce positive payoffs, allowing a higher payoff from one path to be offset by a lower payoff from the other. For higher strikes, zero payoffs become more frequent, which may reduce the negative correlation between paired payoffs and limit the effectiveness

of AV. This provides a possible explanation for the observed results, although the pair correlations are not directly measured. The estimated VRF below one is discussed in Section 5.2.1.

For MC+IS at $d = 64$, measured variance falls at every tested strike, with VRFs ranging from 4.01 to 6.55. MC+IS also reduces measured variance for every tested contract at $d = 16$ and $d = 128$. At each monitoring frequency, the largest gain occurs at $K = 120$, although gains are not monotonic in strike. Shifting the normal inputs towards higher asset paths can make positive payoffs more frequent, with likelihood-ratio weighting preserving the pricing expectation. These results use the independently pilot-selected shift held fixed during evaluation. The pilot diagnostics are examined in Section 5.2.2, while the additional computational cost is discussed in Section 5.4.

For RQMC+AV at $d = 64$, VRFs are 0.89, 0.66, 1.93 and 1.11 for $K = 90, 100, 110, 120$, respectively. Unlike MC+AV, there is no consistent improvement or clear trend across strikes. Results at $d = 16$ and $d = 128$ likewise show that antithetics do not uniformly improve RQMC. At a fixed payoff budget, any benefit from pair averaging must compensate for using half as many Sobol' base points. Section 5.2.1 discusses this distinction.

For RQMC+IS at $d = 64$, measured variance increases at $K = 90, 100, 110$, with VRFs of 0.47, 0.42 and 0.60, but falls at $K = 120$, where the VRF is 3.19. The same contrast appears at $d = 16$ and $d = 128$: VRFs are below one at $K = 90$ and $K = 100$, but above one at $K = 120$. Thus, the pilot-selected shifts benefit RQMC for the highest-strike contracts tested, while shifts that improve MC can worsen RQMC. Section 5.2.2 relates this contrast to the pilot objective and dependence between evaluations within a scrambled Sobol' net.

5.2.1 Interpreting the antithetic-variance results

The MC+AV factor of 0.85 at $K = 110, d = 64$ requires care. For the monotone payoff considered here, (15) implies that the population variance ratio at matched payoff cost is

$$\mathrm{VRF}_{\mathrm{MC+AV}} = \frac{\sigma_Y^2}{\sigma_Y^2 + c_d} = \frac{1}{1 + \rho_d} \geq 1, \qquad \rho_d = \frac{c_d}{\sigma_Y^2}, \tag{24}$$

provided the payoff and pair-average variances are positive. This means that MC antithetics cannot increase the true estimator variance for this payoff at a matched payoff budget.

However, the reported VRFs use variances estimated from separate sets of only 32 replicates per method. At $K = 110, d = 64$, the observed VRFs are 0.90, 2.15 and 0.85 as the evaluation budget increases. Sampling variation can produce an estimated VRF below one despite the theoretical guarantee. These results leave the size of the variance reduction uncertain.

The RQMC comparison has no corresponding guarantee, because its within-net dependence and base-net size also affect estimator variance. As described in Section 3.6, the comparison at matched payoff budgets uses B base points for the original integrand and $B/2$ base points for the pair-averaged integrand.

5.2.2 Interpreting the importance-sampling results

At $d = 64$, the selected scalar shift rises from 0.5 at $K = 90$ to 1 at $K = 100$ and 1.5 at $K = 110$ and $K = 120$. The pilot selects shifts from the finite grid specified in Section 4.3, so they are not necessarily globally optimal.

The saved pilot diagnostics provide additional evidence about the IS results. Table 5 compares the unshifted and selected candidates using each contract's 4,096 common pilot inputs. Positive counts are diagnostic observations from the pilot, rather than additional independent evaluation samples. The second moments refer to the weighted, discounted payoff, so they account for both the changed paths and the likelihood ratio.

Table 5: IS pilot diagnostics at $d = 64$. Counts are positive weighted payoffs out of 4,096. $\widehat{m}_2(a)$ is the estimated second moment in the pilot objective, and $\widehat{a}$ is the selected shift.

K	$\widehat{a}$	Positive, $a = 0$	Positive, $\widehat{a}$	$\widehat{m}_2(0)$	$\widehat{m}_2(\widehat{a})$
90	0.5	3484	3786	262.778	200.621
100	1	2246	3457	95.353	53.191
110	1.5	1081	3036	30.950	10.558
120	1.5	321	1964	7.283	1.231

At strike 120, the zero-shift pilot gives 321 positive payoffs, about 7.8% of its inputs. The selected shift raises that count to 1,964, about 47.9%, and lowers the estimated weighted second moment from 7.283 to 1.231. More pilot inputs reach the positive payoff region while the estimated weighted second moment falls, supporting the intended mechanism. However, selecting the largest shift would not be better merely because it produces more positive payoffs. For the same contract, $a = 3$ gives 3,699 positive payoffs but a larger estimated second moment of 3.893. The likelihood weights make both the size and variability of the weighted contributions relevant to efficiency.

These pilot improvements target IID sampling. For a fixed shift and N dependent RQMC weighted payoffs $H_1, \ldots, H_N$, the variance identity

$$\mathrm{Var}\left(\frac{1}{N}\sum_{i=1}^{N} H_i\right) = \frac{1}{N^2}\left\{\sum_{i=1}^{N}\mathrm{Var}(H_i) + 2\sum_{i<j}\mathrm{Cov}(H_i, H_j)\right\} \tag{25}$$

shows why the pilot objective is insufficient to predict RQMC gains. Changing the weighted integrand can change the cross-covariance terms as well as the individual-payoff variance. The experiment does not estimate these terms separately, so this identity explains why IID variance reductions need not carry over to RQMC rather than proving a particular cause for each observed loss. The contrast between the lower-strike RQMC+IS factors and the gain at $K = 120$ is consistent with a contract-dependent interaction between the transformation and the scrambled-net estimator. He et al. (2023) demonstrate a related effect using Gaussian Laplace proposals. Their covariance-contraction examples do not directly explain the present losses, since this study shifts the mean while retaining the identity covariance. The analysis of Chen et al. (2025) also examines integrand regularity and growth conditions for RQMC-IS convergence. Those asymptotic results do not establish a finite-budget variance reduction for the pilot-selected shifts tested here.

5.3 Monitoring frequency and its interaction with moneyness

Figure 3 compares performance across $d = 16, 64, 128$ monitoring dates at $K = 90, 100, 120$, using $B = 16\,384$ and $R = 32$. Baseline RQMC has lower measured RMSE than baseline MC in all nine configurations. Its relative advantage over MC is smaller at 128 dates than at 16 dates for each strike, although the change across the three schedules is not

always monotonic. Absolute RQMC errors also vary non-monotonically. For example, at $K = 100$, RMSE is approximately 0.00525, 0.00974 and 0.00740 as d increases.

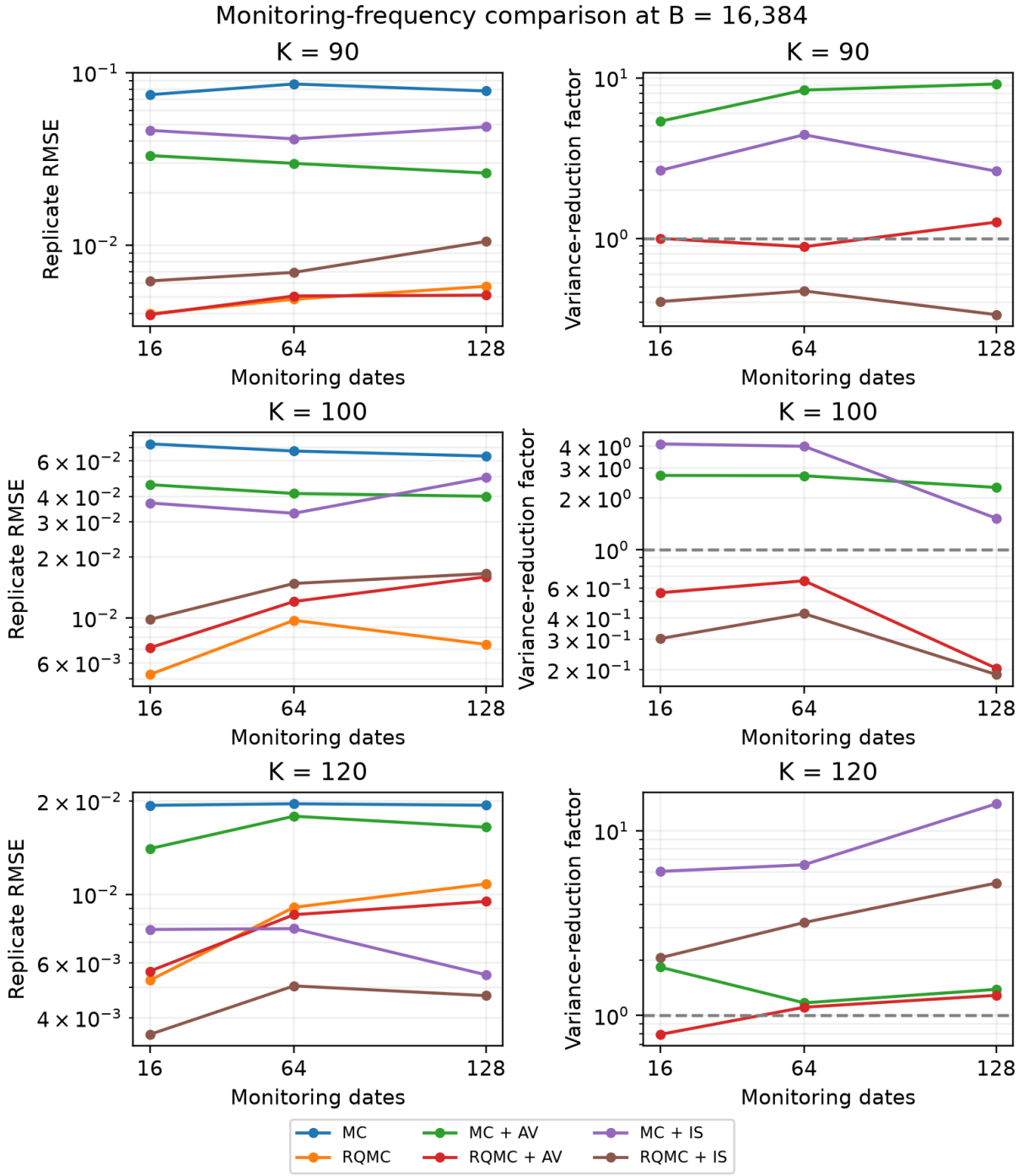


Figure 3: RMSE on the left and VRFs on the right across $d = 16, 64, 128$ monitoring dates at $K = 90, 100, 120$, with $B = 16\,384$ and $R = 32$. Each monitoring schedule uses its own contract-specific benchmark. Dashed lines on the right mark a VRF of one.

At $K = 90$, the measured benefit of MC+AV increases with monitoring frequency. Its VRF rises from 5.36 to 8.39 and 9.15 across the three schedules. RQMC+AV VRFs remain much closer to one, indicating that the stronger MC gains do not carry over consistently to RQMC.

At $K = 100$, MC+AV provides a relatively stable improvement, with VRFs between 2.31 and 2.71. In contrast, both AV and IS increase measured RQMC variance at every monitoring frequency. The largest proportional increases occur at $d = 128$, where the RQMC+AV and RQMC+IS VRFs are approximately 0.20 and 0.19, respectively.

At $K = 120$, IS reduces measured variance for both samplers at every monitoring frequency. MC+IS VRFs are 6.03, 6.55 and 14.00, while RQMC+IS VRFs are 2.05, 3.19 and 5.21 for $d = 16, 64, 128$, respectively. Thus, the measured IS gains increase with monitoring frequency for these highest-strike contracts.

These patterns show that the effect of monitoring frequency depends on both moneyness and the method used. They do not isolate the effect of integration dimension. Changing the monitoring schedule also changes the contract and payoff distribution, with each schedule assessed against its own benchmark. For the $K = 120$ IS comparisons, the selected shift also changes from 2 at $d = 16$ to 1.5 at $d = 64$ and $d = 128$. The increasing VRFs therefore cannot be attributed solely to higher dimension.

5.4 Runtime and the cost of selecting an IS shift

Figure 4 compares RMSE against runtime across strikes at $d = 64$. For MC, RQMC and their AV variants, runtime is the mean evaluation time per replicate. For IS, the standalone convention adds one full pilot time to the mean evaluation time, representing the cost of selecting a shift and then producing one price estimate. The experiment itself reuses one selected shift per contract across both IS methods, all budgets and all evaluation replicates, as described in Section 4.3.

At $K = 120, d = 64, B = 16\,384$, RQMC+IS achieves an RMSE of 0.00506, compared with 0.00908 for baseline RQMC. Its standalone runtime is approximately 0.0666 seconds, including tuning, compared with 0.0414 seconds for baseline RQMC. This demonstrates greater accuracy at a higher computational cost for this configuration. Figure C.2 in Appendix C separates the evaluation and pilot times for this comparison.

The relative overhead of tuning decreases as the evaluation budget increases. The pilot requires 28,672 payoff evaluations, equivalent to 28 times the evaluation budget at $B = 1\,024$, but only 1.75 times at $B = 16\,384$. These are payoff-count ratios, rather than runtime ratios, which may differ due to the costs of input generation and candidate evaluation. Reusing a selected shift across repeated valuations of the same contract would spread the pilot cost over those valuations, but the results here retain the standalone convention.

AV also affects runtime because it requires only half as many base input vectors at a matched payoff budget, while evaluating two payoffs per vector. Equal payoff counts therefore do not imply equal elapsed time, and the VRFs in Section 5.2 must be considered alongside runtime. Since the experiments use fixed payoff budgets rather than fixed runtimes, they do not establish which method achieves the lowest RMSE at equal runtime.

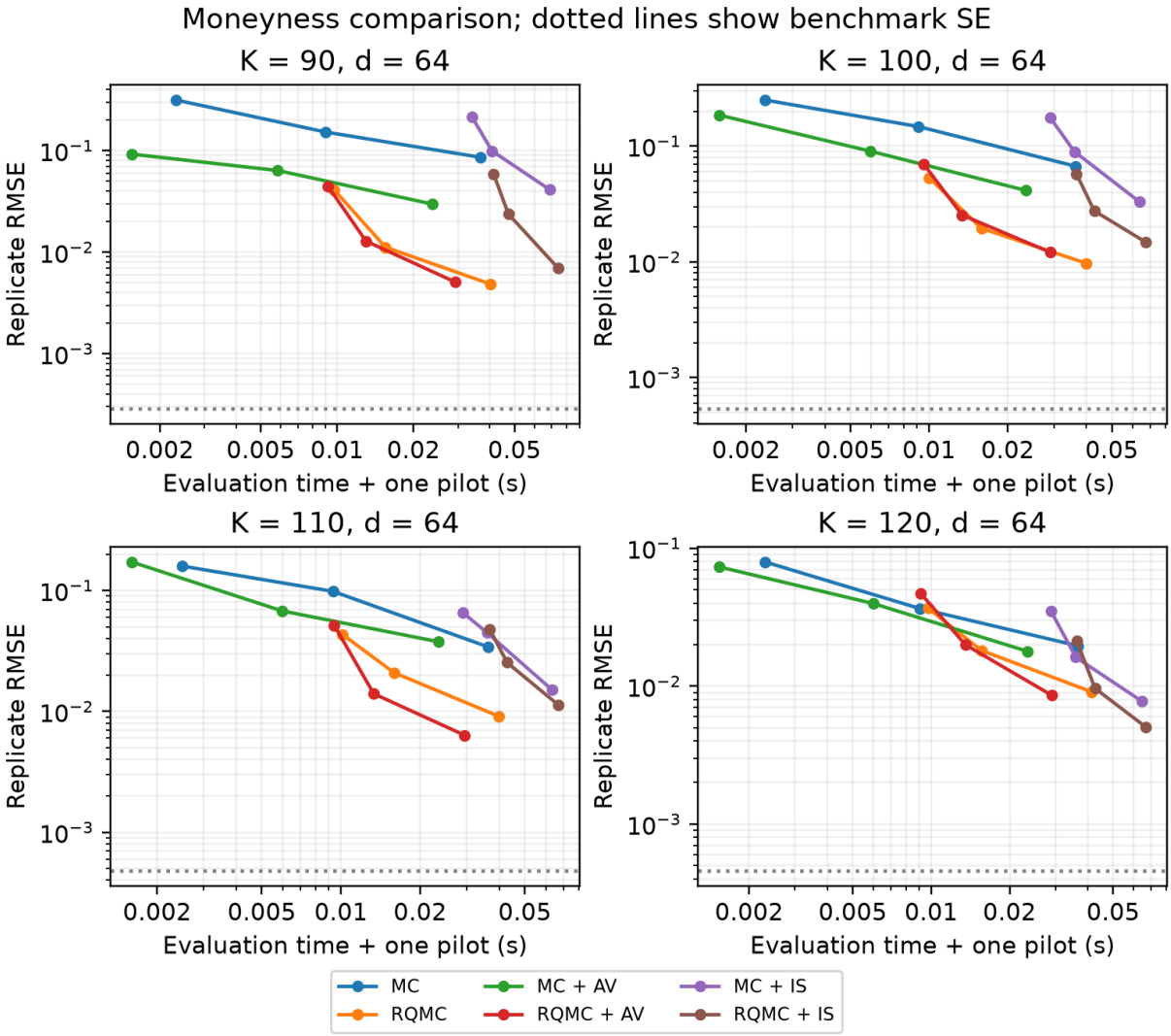


Figure 4: RMSE against standalone runtime across strikes at $d = 64$ and $R = 32$. Runtime is the mean evaluation time per replicate, with one full pilot time added for IS. IS accuracy is conditional on the selected shift. Each curve contains results at $B = 1\,024, 4\,096, 16\,384$.

5.5 Limitations

The findings are limited to arithmetic Asian calls under the tested Black–Scholes parameters, using a forward-increment path construction and a finite range of evaluation budgets. They therefore do not establish a general ranking across other contracts, market parameters or path constructions. Changing monitoring frequency also changes the contract and payoff distribution, so the dimension comparisons do not isolate the effect of integration dimension.

Each comparison uses 32 independent evaluation replicates, leaving estimated VRFs near one uncertain. The MC+AV result discussed in Section 5.2.1 illustrates how sampling variation can produce an estimated VRF below one despite a theoretical guarantee of no variance increase. The comparisons are descriptive and do not establish statistically significant differences across all tested configurations. The three evaluation budgets also provide insufficient evidence to determine asymptotic convergence rates. Numerical benchmarks have estimated, nonzero uncertainty and share the pricing implementation with the evaluated estimators. Independent analytical checks support implementation correctness but cannot rule out all shared errors.

IS performance is assessed conditional on one pilot-selected shift per contract. The finite candidate grid and IID second-moment objective limit the shifts considered and do not directly target RQMC estimator variance. The results therefore do not establish how performance would vary under repeated pilot selection or a tuning procedure designed for RQMC. Runtime comparisons include standalone pilot costs but remain specific to the implementation and execution conditions. A separate experiment at equal total computational cost would be needed to compare the complete tuning-and-evaluation procedures on that basis.

6 Conclusion

The study provides empirical answers to the four research questions for the tested arithmetic Asian calls under Black–Scholes:

1. **MC versus QMC.** Scrambled Sobol' RQMC gives smaller measured errors than standard MC at matched evaluation budgets throughout the tested grid. At 64 dates, its largest-budget RMSE advantage ranges from about 2.2 to 17.7 times across the four strikes.

2. **Monitoring frequency.** The RQMC advantage persists at 16, 64 and 128 dates, but its magnitude changes with the schedule and is not uniformly monotonic. Each comparison uses its own contract reference.

3. **Variance reduction with QMC.** AV and IS do not consistently improve RQMC, even when the same technique helps MC. IS provides a clear measured gain for the strike-120 contracts, but increases RQMC variance at lower strikes at the largest budget (VRF below one).

4. **Dependence on option characteristics.** MC antithetics are particularly effective for the strike-90 contracts, while IS is useful for the out-of-the-money strike-120 contracts under both samplers. The size and sometimes the direction of improvement depend on both moneyness and monitoring frequency.

RQMC provides a strong baseline for the arithmetic Asian calls tested, while the additional value of AV and IS depends on moneyness, monitoring frequency and computational cost. Further work could use more independent replicates to clarify comparisons with VRFs near one and repeat the full pilot-and-evaluation procedure to assess IS performance at equal total computational cost. A further direction is to select shifts using estimated variance across independent scrambled Sobol' replicates, with separate evaluation samples and explicit accounting for tuning cost. This would examine whether tuning specifically for RQMC improves upon the IID pilot used here. The findings support evaluating variance-reduction techniques against each sampler's own baseline, since improvements under standard MC do not necessarily translate into improvements under RQMC.

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A Experiment Algorithm

This algorithm summarises the sequencing of the main study. It uses the estimators in Section 3 and the parameter choices in Section 4.1. The analytical validation checks are performed separately before interpreting the resulting comparisons.
Inputs. Contract configurations, six methods, evaluation budgets B, replication count R, reference size and replication count, and the IS pilot size and candidate-shift grid.
Outputs. Replicate price estimates and timings, numerical references and their standard errors, selected shifts and pilot diagnostics, and comparison summaries at each contract and budget.

1. **Form the contract list.** Combine the moneyness and monitoring-frequency grids and remove overlapping configurations. Process the ten resulting contracts once each. Reserve separate random streams for all pilots, reference replicates and evaluation replicates.

2. **Select the IS shift for the current contract.** Generate the independent MC pilot inputs. For each candidate a, calculate $h_{d,\boldsymbol{\theta}(a)}$ from (19), its sample second moment and its positive-payoff count. Apply (22), exclude uninformative candidates, and retain the documented zero-shift fallback if needed. Fix $\widehat{a}$ for both IS methods and record the total pilot cost.

3. **Construct the reference.** Generate the specified number of independent baseline RQMC estimates using the larger reference net. Average these estimates and calculate their across-scramble standard error. All reference streams are separate from the pilot and evaluations.

4. **Evaluate each budget, replicate and method.** For AV, set $N = B/2$; otherwise set $N = B$. Generate either IID MC uniforms or a complete scrambled Sobol' net of N points, using a fresh stream for each estimate. Apply the common endpoint convention and inverse-normal transform. Calculate $g_d(\boldsymbol{Z}_i)$ for a baseline, $[g_d(\boldsymbol{Z}_i) + g_d(-\boldsymbol{Z}_i)]/2$ for AV, or $h_{d,\boldsymbol{\theta}(\widehat{a})}(\boldsymbol{Z}_i)$ for IS. Average the contributions and record the estimate, runtime, input and payoff counts, shift and endpoint adjustments. Reject numerical failures explicitly. Any within-replicate MC standard error uses independent contributions, including pair averages for AV.

5. **Summarise independent replicates.** At each contract and budget, calculate the mean, estimator variance, standard error of the mean and benchmark-relative errors using (11)–(13). Compare each modified method with its corresponding baseline using (23). Reuse the same baseline rows for AV and IS; do not replace across-scramble uncertainty with an IID within-net formula.

6. **Check reference precision.** Compare reference SE with 10% of each measured RMSE. If precision is inadequate, strengthen that reference with independent samples or limit the interpretation. In this study, 48 reference replicates were appended for $K = 120, d = 128$, leaving the original evaluation estimates unchanged. Recalculate reference-dependent summaries after the refinement.

7. **Compare accuracy and cost.** Report variance reduction at matched evaluation budgets and show runtime separately. For a standalone IS valuation, add one full pilot to one evaluation; do not sum this hypothetical charge over the actual

experiment, which shares the pilot. Generate the figures and tables from the saved results.

The project repository is available at https://github.com/eugene-xiang/fyp-mc-vs-qmc-option-pricing. Instructions for running the experiments and reproducing the figures and tables are provided in the repository's README.

B Additional Sweep Results

Table 6 gives the errors underlying the largest-budget comparisons in Section 5. It uses the main sweep and the same saved baseline estimates.

Table 6: Replicate RMSE at $B = 16\,384$, with 32 independent estimates per method and contract. Prices and errors are in the same units as $S_0 = 100$.

K	d	MC	RQMC	MC+AV	RQMC+AV	MC+IS	RQMC+IS
90	16	0.07450	0.00399	0.03305	0.00395	0.04630	0.00619
100	16	0.07278	0.00525	0.04571	0.00711	0.03710	0.00983
120	16	0.01936	0.00528	0.01403	0.00565	0.00770	0.00353
90	64	0.08597	0.00485	0.02977	0.00507	0.04131	0.00694
100	64	0.06707	0.00974	0.04134	0.01206	0.03298	0.01481
110	64	0.03418	0.00908	0.03773	0.00635	0.01507	0.01134
120	64	0.01959	0.00908	0.01786	0.00859	0.00775	0.00506
90	128	0.07828	0.00577	0.02616	0.00512	0.04850	0.01049
100	128	0.06331	0.00740	0.04002	0.01599	0.04956	0.01660
120	128	0.01937	0.01080	0.01646	0.00948	0.00550	0.00471

C Supplementary Graphical Results

Both figures use the main-sweep contract with $S_0 = 100$, $K = 120$, $T = 1$, $r = 0.05$, $\sigma = 0.20$ and $d = 64$. Each method uses $B = 16\,384$ payoff evaluations per replicate and $R = 32$ independent replicates. The IS shift is fixed at the pilot-selected value $\widehat{a} = 1.5$.

C.1 Mean prices and reference uncertainty

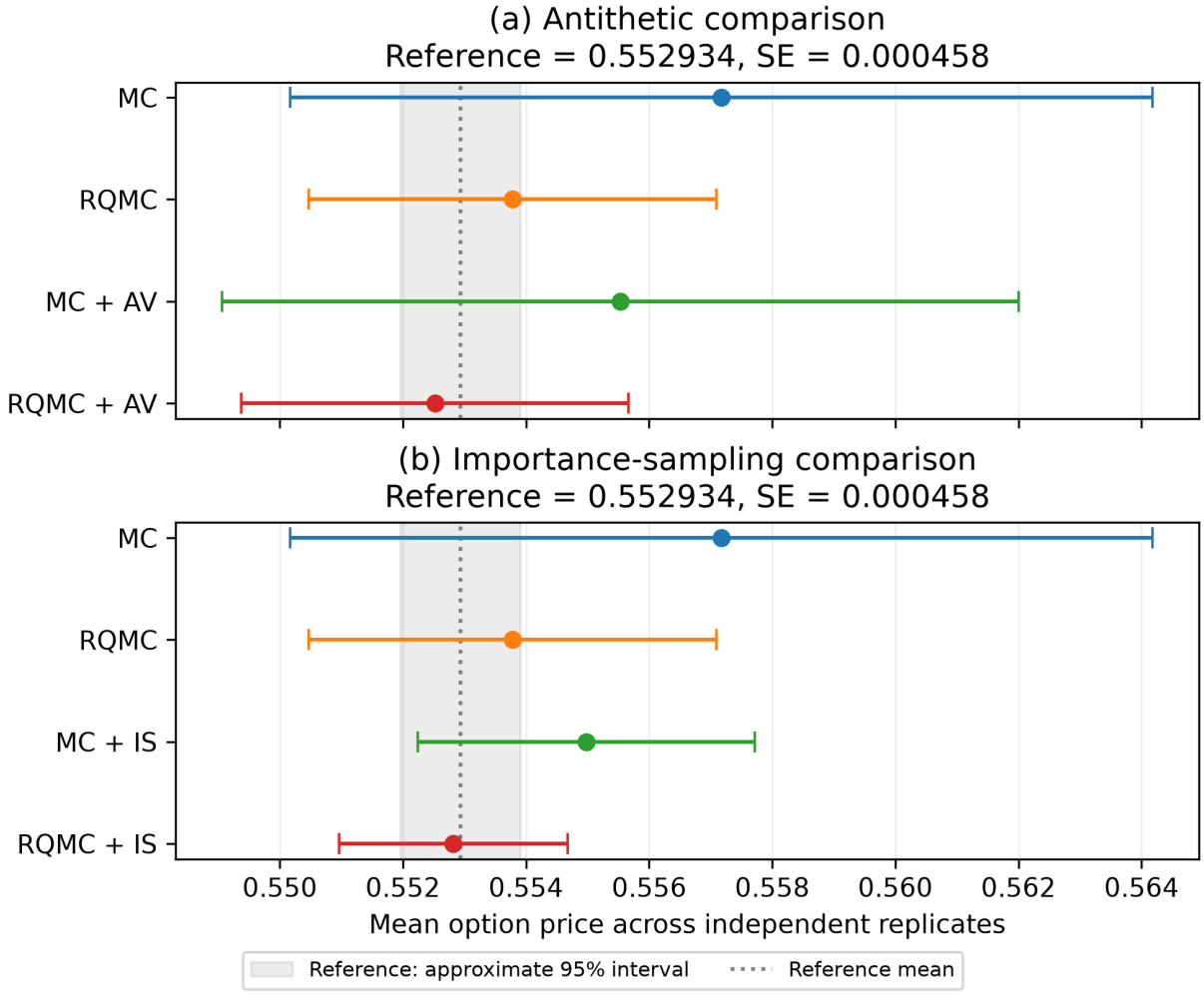


Figure C.1: Mean prices and approximate 95% Student-t intervals. Dots are replicate means, with intervals $\overline{V} \pm t_{0.975,31}\,\mathrm{SE}(\overline{V})$. The vertical line is the numerical reference, and the shaded band is $V^{\mathrm{ref}} \pm t_{0.975,15}\,\mathrm{SE}(V^{\mathrm{ref}})$ from 16 independent reference scrambles. Both panels share the same baseline estimates. IS intervals condition on the selected shift.

The intervals describe uncertainty in the mean of 32 replicates rather than the error of an individual run. The RQMC intervals use independent scrambles and have approximate coverage (L'Ecuyer et al., 2023).

C.2 Evaluation and importance-sampling tuning costs

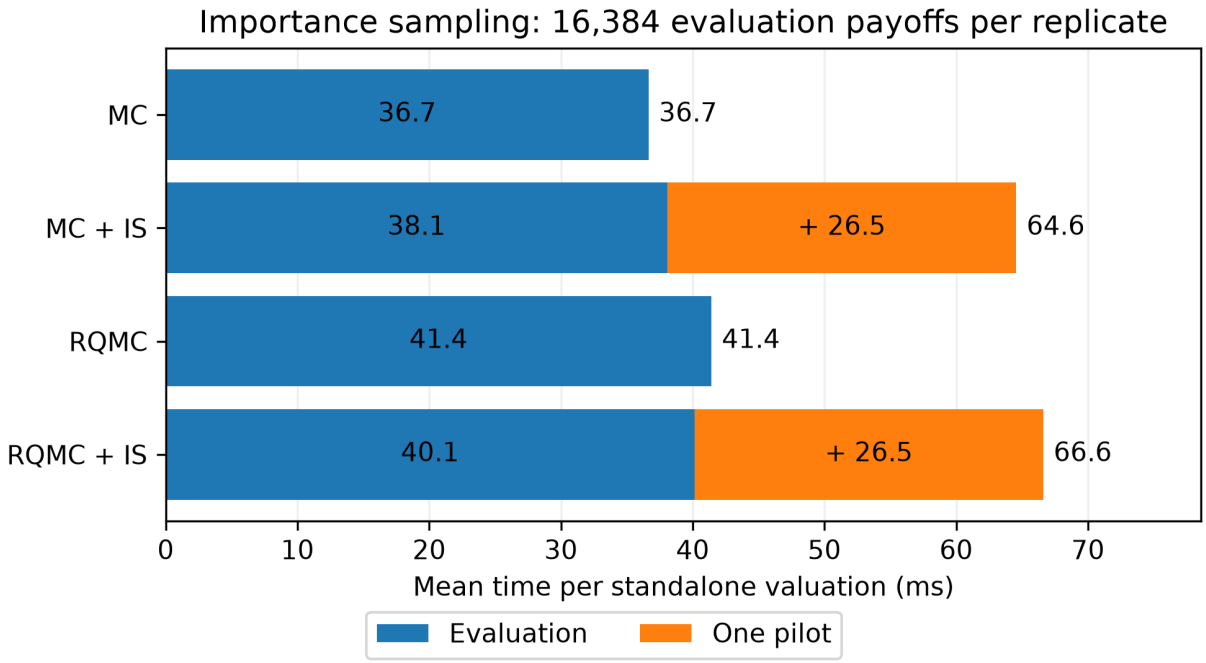


Figure C.2: Mean evaluation time and the additional pilot cost for a standalone valuation. Each IS bar adds one pilot of approximately 26.5 ms and 28,672 payoff evaluations, giving 45,056 total payoffs compared with 16,384 for the baseline. Evaluation times are averaged over 32 replicates. Labels are in milliseconds and rounded independently.

The actual experiment shares one pilot across both IS methods and all budgets and replicates. The bars show standalone costs under the convention in Section 4.3.

D Supplementary Implementation Checks

The following checks support the mathematical validation in Section 4.4:

- **Sampling and numerical safeguards.** A complete scrambled Sobol' net has the expected one-dimensional stratification. Non-power-of-two Sobol' base counts are rejected, including for antithetic sampling. Endpoint adjustments count clipped base uniform coordinates, while antithetic normal vectors are generated by negation without a second inverse normal transformation. RQMC standard errors are calculated across independent scrambles rather than within individual nets. Invalid inputs, non-finite calculations and numerical underflow are rejected.

- **Reproducibility and budget accounting.** Small repeated experiments reproduce estimates, seeds and payoff counts. Baselines use B base vectors, and antithetic methods use $B/2$ vectors and $B/2$ pairs at the same payoff budget. Reference and evaluation seeds are distinct. VRFs are calculated relative to the corresponding baseline at the same budget, with ratios involving zero denominators reported as undefined.

- **Pilot separation and safeguards.** Selection excludes candidates with no positive pilot payoffs and flags a zero-shift fallback if every candidate is uninformative. Checks confirm reproducible selection, separation of pilot and evaluation seeds, recorded shifts, and tuning-payoff and runtime accounting.

- **Sweep orchestration.** A small repeated sweep checks that overlapping stages reuse each contract, random streams remain distinct across all contracts and phases, and saved replicate estimates reproduce the reported RMSE against each contract's own benchmark.